

# MUNAM AHMAD

CIVIL ENGINEER | TRANSPORTATION | CONSTRUCTION PROJECT MANAGEMENT

+92 333 8904383 | [munamnoor0711@gmail.com](mailto:munamnoor0711@gmail.com) | <https://www.linkedin.com/in/munam-ahmad-989118330/> | Islamabad, Pakistan

## PROFESSIONAL SUMMARY

I am a dedicated and detail oriented civil engineer recently graduated from NUST. With a strong foundation in site engineering and project management, I have engaged in various projects that emphasize cost analysis, scheduling, and innovative solutions in construction. I possess practical expertise in cost estimation, quantity surveying, and project scheduling using tools such as Primavera P6 and MS Project. My technical proficiency spans industry-standard software including AutoCAD, Civil 3D, and Revit, enabling accurate design, planning, and documentation. I have led and contributed to projects involving traffic analysis, cost estimation, design and my final year design project is based upon construction automation using BIM and robotic technologies. I am driven by innovation and precision and thrive in collaborative, deadline oriented environments with a commitment to safety and quality.

## PROJECTS

### Cost Analysis and Scheduling of a Road Construction Project near NUST

Nov 2024 - Dec 2024

- Performed detailed analysis and calculations for a 1.8 km two-way road construction project, designed by Nasir International Private Limited, including equipment productivity and haul distances.
- Completed comprehensive cost estimations covering machinery, material quantities, and labor requirements.
- Managed the full scheduling of 18 construction activities, within a time duration of 2 months, and coordinated resource allocation for efficient project execution.

### Road Traffic Analysis and Intersection Design

Apr 2024 - May 2024

- Conducted traffic data analysis using **Synchro Software** to evaluate intersection performance and signal timings.
- Designed and optimized intersection layouts using **SIDRA Intersection software**, enhancing traffic flow and safety.

### Design of a Piping Rack System in NASPHA LPG Plant

Nov 2024 - Dec 2024

- Designed a **piping rack system** considering layout optimization, load conditions, and safety standards for an LPG processing facility.
- Applied relevant piping codes and standards to ensure compliance and operational integrity.
- Collaborated on routing strategies to minimize material usage and support system efficiency within the plant.

### Design of a Robotic Arm for Wall Construction (FYDP)

Sep 2024 - Sep 2025

- Researched and developed a semi-automated robotic arm system aimed at advancing construction automation.
- Integrated Building Information Modelling (BIM) for real-time automation in masonry wall construction.
- Compared traditional manual methods with automated robotic processes to evaluate efficiency gains.
- Focused on addressing labor shortages and enhancing productivity in the construction industry through innovation.

### Quantity Surveying and Cost Estimation of Zaman Villas, Bahawalpur

Apr 2024 - May 2024

- Conducted detailed quantity takeoffs for all structural and architectural components of a ( **55x35 ft** ) residential villa.
- Estimated the quantities and cost for ground floor containing 1 bedroom, 1 lounge, 1 drawing room, 1 mater kitchen, 1 porch, 3 bathrooms and a lawn.
- Prepared comprehensive cost estimates covering materials, labor, and overheads based on market rates and project specifications.
- Performed complete calculations from excavation to substructure and superstructure, including foundations, columns, beams, and slabs.
- Developed a **Bill of Quantities (BOQ)** and assisted in budgeting and financial planning for the construction phase.
- Ensured accuracy and consistency by cross-verifying drawings, site conditions, and contractor inputs.

## PROFESSIONAL EXPERIENCE

### Internee Engineer | PMO, NUST

Jun 2024 - Aug 2024

- Performed Site engineering internship with PMO on the construction site of NBS building at NUST, H-12.
- Gained hands-on experience in site engineering by assisting with on-ground construction activities and quality checks.
- Contributed to project scheduling using construction timelines and progress tracking tools to ensure timely completion.
- Participated in the procurement process by helping with material requisitions, vendor coordination, and inventory checks.
- Supported construction supervision through daily site inspections, specifically involving the knowledge of BBS, flooring, and staircases, and reporting to ensure adherence to design specifications and safety protocols.
- Applied problem-solving skills to address on-site challenges by coordinating with engineers and labor team

EDUCATION

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|---------------------------------------------------------------------|----------------------------|
| <b>Bachelor of Civil Engineering</b><br>NUST, Islamabad             | <b>Sep 2021 - Present</b>  |
| <b>F.Sc Pre-Engineering</b><br>Govt. Post Graduate College, Bhakkar | <b>Jun 2017 - Jun 2019</b> |

CERTIFICATIONS AND PUBLICATIONS

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**Certificate of Participation – Civil 3D Hands-On Training Workshop**

- Successfully participated in a professional workshop focused on geometric design of highways using **Autodesk Civil 3D**.
- Gained hands-on experience with road alignment, profile design, corridor modeling, and intersection layout in Civil 3D.
- Strengthened understanding of transportation design principles and their implementation in a real-world software environment.
- Interacted with faculty and professionals from the School of Civil and Environmental Engineering (SCEE), NUST.

**Conference Papers**

**Comparative Analysis of Cloverleaf and Diverging Diamond Interchanges: Design, Efficiency, and Safety**

4th International Conference on Frontiers in Academic Research December 13-14, 2024: Konya, Turkey

- Presented research paper titled "**Comparative Analysis of Cloverleaf and Diverging Diamond Interchanges: Design, Efficiency, and Safety**" at the 4th International Conference on Frontiers in Academic Research, Konya, Turkey Delivered
- oral presentation on interchange design evaluation, emphasizing safety and operational efficiency in transportation infrastructure.
- Engaged in international academic discourse and networking with researchers in the field of civil/transportation engineering.

RELEVANT SKILLS

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Autodesk Revit | AutoCAD | Civil 3D | Revit | Synchro | SolidWorks | Microsurvey CAD | SIDRA Intersections | Primaverra P6 | Problem Solving | MS Office | Site Supervision | Cost Estimation | Quantity Surveying

ADDITIONAL INFORMATION

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**Languages**

- English, Urdu, Punjabi, Saraiki